

Solution

SOUND-01

Class 09 - Science

Section A

- Potential Energy:** It is the energy possessed by a body by virtue of its position or shape.
- In a toy car, the wound spring possesses potential energy. When spring is released, its potential energy changes into kinetic energy due to which the toy car moves.
- Given, $m = 50 \text{ kg}$, $h_1 = 75 \text{ m}$, $h_2 = 60 \text{ m}$
At point A, $PE_1 = mgh_1 = 50 \times 10 \times 75$
 $= 37500 \text{ J}$
At point B, $PE_2 = mgh_2 = 50 \times 10 \times 60$
 $= 30000 \text{ J}$
Change in PE = $PE_1 - PE_2 = 37500 - 30000 = 7500 \text{ J}$
- The kinetic energy of an object of mass m moving with a velocity v is given by the expression $\frac{1}{2}mv^2$. To bring the object to rest, an equal amount of work i.e. $\frac{1}{2}mv^2$ is required to be done on the object.
- Power is defined as the rate of doing work.
$$\text{Power} = \frac{\text{Work done}}{\text{Time taken}}$$
$$P = \frac{W}{t}$$
- The SI unit of power is watt which is denoted by symbol W.
1 watt is the power of an appliance which does work at the rate of 1 joule per second.
- The power of an electrical appliance tells us the rate at which electrical energy is consumed by it.
- 1 kilowatt = 1000 watts
 $1 \text{ kW} = 1000 \text{ W}$
1 megawatt = 1000,000 watts
 $1 \text{ MW} = 10^6 \text{ W}$
- When a force of 1 Newton causes a displacement of 1 m in its own direction, the work done is said to be one joule.
- $p = 5000 \text{ W}$, $t = 2 \text{ h}$
Energy consumed = $pt = 5000 \times 2 = 10000 \text{ Wh} = 10 \text{ kW} = 10 \text{ units}$.
- When a carpet is beaten with a stick, the dust comes out of it because of the law of inertia. Initially, the dust particles are at rest along with the carpet. Beating the carpet with the stick makes the carpet move but the dust particles remain at rest due to inertia at rest, thus the dust gets detached from the carpet.
- Given $W = 1000 \text{ J}$, $t = 10 \text{ s}$, $P = ?$
We know, $P = \frac{W}{t} = \frac{1000}{10} = 100 \text{ W}$
- When it strikes ground, its PE is zero and after bouncing, its potential energy increases gradually.
- At the time it strikes the ground, it has maximum KE and after it bounces, its KE starts changing into potential energy.
- The ultimate or total energy remains constant at any point of time during the motion.
- It does not violate the law of conservation of energy, because sum of potential energy and kinetic energy remains always the same.
- Work done on an object is defined as the magnitude of the force multiplied by the distance moved by the object in the direction of the applied force.
Work done = force $\times$ distance
 $W = F \times s$
- $F = 250 \text{ kg} \times 10 \text{ ms}^{-2}$ ($g = 10 \text{ ms}^{-2}$) = 2500 N s = 1 m
 $W = F.s = 2500 \text{ N} \times 1 \text{ m} = 2500 \text{ Nm} = 2500 \text{ J}$
- Zero, as the box does not move at all while holding it.
- In order to hold the box, men are applying a force that is opposite and equal to the gravitational force acting on the box. While applying the force, muscular effort is involved. So, they feel tired.
- Mass = $m = 5 \text{ Kg}$
Height = $h = 12 \text{ m}$.
 $g = \text{Acceleration due to gravity} = 9.8 \text{ m/s}^2$

$$\begin{aligned} \text{P.E.} &= mgh \\ &= 5 \times 12 \times 9.8 \\ &= 588 \text{ J} \end{aligned}$$

22. Mass = $m = 5 \text{ Kg}$

Height = $h = 12 \text{ m.}$

$g = \text{Acceleration due to gravity} = 9.8 \text{ m/s}^2$

Force = ?

Work Done = P. E. energy of the Body

Force $\times$ Distance Moved = 588

$F \times 12 = 588$

$F = 49 \text{ N}$

23. The work done in lifting the body gets stored as potential energy.

24. The work done by a force on a body depends on two factors:

i. Magnitude of force applied.

ii. Displacement in the direction of force applied.

25. **Law of conservation of energy:** The energy in a system can neither be created nor destroyed. It may be transformed from one form to another, but the total energy of the system remains constant.

26. The energy possessed by a body due to a displacement caused in it by the application of a force is called mechanical energy.

27. Energy = Power $\times$ Time = $4 \times 600 \text{ W} \times 5 \text{ h} = \frac{4 \times 600 \times 5}{1000} = \frac{12000}{1000} = 12 \text{ KWh}$

Hence the energy in kWh consumed is 12 kWh.

28. Yes, a body can have energy without possessing momentum. A body may have potential energy and still can be at rest. This means it would have zero momentum and still it could possess energy.

29. Given, mass of the object, $m = 10 \text{ kg}$

Height, $h = 50 \text{ cm} = 0.5 \text{ m}$

As potential energy is given by $PE = mgh$

$= 10 \times 10 \times 0.5$

$= 50 \text{ J}$

30. Given, mass of the object, $m = 10 \text{ kg}$

Height, $h = 50 \text{ cm} = 0.5 \text{ m}$

From the law of conservation of energy, the total energy of the ball just before dropping = total energy of the ball just on touching the ground

$\Rightarrow KE + PE \text{ of the ball just before dropping}$

$\Rightarrow KE = 50 \text{ J}$

31. Given, mass of the object, $m = 10 \text{ kg}$

Height, $h = 50 \text{ cm} = 0.5 \text{ m}$

As we know, $KE = 50 \text{ J}$

So, $\frac{1}{2}mv^2 = 50 \Rightarrow v^2 = \frac{50 \times 2}{10} = 10$

So, the velocity with which it hits the ground,

$v = \sqrt{10} = 3.16 \text{ ms}^{-1}$

32. Given, mass of the object, $m = 10 \text{ kg}$

Height, $h = 50 \text{ cm} = 0.5 \text{ m}$

Yes, a body can have energy without possessing momentum. A body may have potential energy and still can be at rest. This means it would have zero momentum and still it could possess energy.

33. Kinetic energy = $\frac{1}{2}mv^2$

$= \frac{1}{2} \times 900 \times (30)^2 = \frac{1}{2} \times 900 \times 900$

$= 4,05,000 \text{ J}$

34. Energy used = Work done against resistive force

$= \text{Force} \times \text{Distance}$

$= 2,000 \times 30 = 60,000 \text{ J} = 60 \text{ kJ}$

A resistive force is one that inhibits or resists the motion of an object. It acts in a direction opposite to any motion or applied force that is trying to move the object.

$$35. \text{ Minimum power} = \frac{\text{Energy used}}{\text{Time taken}}$$

$$= \frac{60,000 \text{ J}}{1 \text{ s}}$$

$$60,000 \text{ W} = 60 \text{ W}$$

36. Chemical energy. The fuel in your car provides chemical potential energy, which is converted to kinetic energy when it's combusted.

37. The potential energy of the car remains the same since PE (=mgh) is independent of velocity.

38. The kinetic energy of the car becomes four times, since $KE = (\frac{1}{2}mv^2)$ is proportional to square of velocity.

39. The momentum of the car will also get doubled since momentum (p = mv) is proportional to velocity.

40. The sum of the kinetic energy and potential energy of an object is called mechanical energy. The total amount of mechanical energy is merely the sum of the potential energy and the kinetic energy.

41. Given, mass of the vehicle, m = 1 tonne = 1000 kg

Initial speed, u = 60 ms⁻¹

Distance between vehicle and the cow, s = 9 m

Final velocity, v = 0

Kinetic Energy is the energy possessed by a body due to its motion. The kinetic energy of an object increases with its speed.

KE of vehicle before applying brakes is given by = $\frac{1}{2} mu^2 = \frac{1}{2} \times 1000 \times 60 \times 60 = 1800000 \text{ J} = 1800 \text{ kJ}$

42. Given, mass of the vehicle, m = 1 tonne = 1000 kg

Initial speed, u = 60 ms⁻¹

Distance between vehicle and the cow, s = 9 m

Final velocity, v = 0

From the third equation of motion,

$$v^2 - u^2 = 2as$$

$$(0)^2 - (60)^2 = 2 \times a \times 9$$

$$\Rightarrow a = -200 \text{ ms}^{-2}$$

So, retarding force provided by the brakes = ma = 1000 × (-200) = -200000 N

43. Given, mass of the vehicle, m = 1 tonne = 1000 kg

Initial speed, u = 60 ms⁻¹

Distance between vehicle and the cow, s = 9 m

Final velocity, v = 0

Now, again from the second equation of motion,

$$s = ut + \frac{1}{2} at^2 \Rightarrow 9 = 60t + \frac{1}{2} \times (-200) t^2$$

$$\text{or } 9 = 60t - 100t^2$$

$$\text{or } 100t^2 - 60t + 9 = 0 \Rightarrow (10t - 3)^2 = 0$$

$$\text{or } 10t - 3 = 0 \Rightarrow t = \frac{3}{10} = 0.3 \text{ s}$$

44. Given, mass of the vehicle, m = 1 tonne = 1000 kg

Initial speed, u = 60 ms⁻¹

Distance between vehicle and the cow, s = 9 m

Final velocity, v = 0

So, work done by the braking force is given by = Fs = -200000 × 9 = -1800000 J = -1800 kJ

Section B

45. Fill in the blanks:

(i) 1. Units

(ii) 1. Negative

(iii) 1. Potential

(iv) 1. Positive

(v) 1. Displacement

(vi) 1. Displacement

(vii) 1. Kinetic

(viii) 1. Friction

- (ix) 1. Watt
- (x) 1. Zero
- (xi) 1. Power
- (xii) 1. 1 joule
- (xiii) 1. Scalar
- (xiv) 1. Kinetic
- (xv) 1. Motion
- (xvi) 1. Watt
- (xvii) 1. Electrical
- (xviii) 1. Kilowatt-hour

Section C

- 46. The correct order of match is given as (a) – (ii), (b) – (iii), (c) – (iv), (d) – (i).
- 47. The correct order of match is given as (a) – (iv), (b) – (iii), (c) – (i), (d) – (ii).
- 48. The correct order of match is given as (a) – (ii), (b) – (iii), (c) – (iv), (d) – (i).
- 49. The correct order of match is given as (a) – (iii), (b) – (iv), (c) – (ii), (d) – (i).
- 50. The correct order of match is given as (a) – (iv), (b) – (iii), (c) – (ii), (d) – (i).
- 51. (a) - (iv), (b) - (iii), (c) - (i), (d) - (ii)
- 52. The correct order of match is given as (a) – (iv), (b) – (iii), (c) – (ii), (d) – (i).
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